DG INTERNS HUB

WEEK 1 CYBERSECURITY INTERNSHIP REPORT

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Program: Cybersecurity Internship
Reporting Period: Week 1
Virtualization Platform: VMware Workstation
Lab Systems: Kali Linux and Metasploitable 2

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1. INTRODUCTION

Cybersecurity is the practice of protecting computers, networks, applications, and data from unauthorized access, attacks, and damage. During Week 1 of the internship, I focused on understanding cybersecurity fundamentals and developing basic practical skills in Linux, networking, virtual machine environments, and security learning platforms.

The work included preparing theory notes, configuring and inspecting a virtual lab, practicing Linux commands, studying network connectivity, completing OverTheWire Bandit Levels 0–10, and working through eight beginner-level TryHackMe rooms.

Screenshots, diagrams, and write-ups were organized into separate folders to maintain evidence of the work performed.

2. OBJECTIVES

The main objectives of Week 1 were:

1. Understand the fundamentals of cybersecurity and the CIA Triad.
2. Learn about SOC, VAPT, and incident response.
3. Understand the OSI and TCP/IP networking models.
4. Practice Linux commands and file permissions.
5. Understand users, groups, shared directories, and scheduled tasks.
6. Learn basic network configuration and connectivity testing.
7. Understand virtual machine networking and safe lab isolation.
8. Complete beginner-level cybersecurity exercises on TryHackMe.
9. Practice Linux command-line challenges through OverTheWire Bandit.
10. Document practical activities and organize technical evidence.

3. VIRTUAL LAB ENVIRONMENT

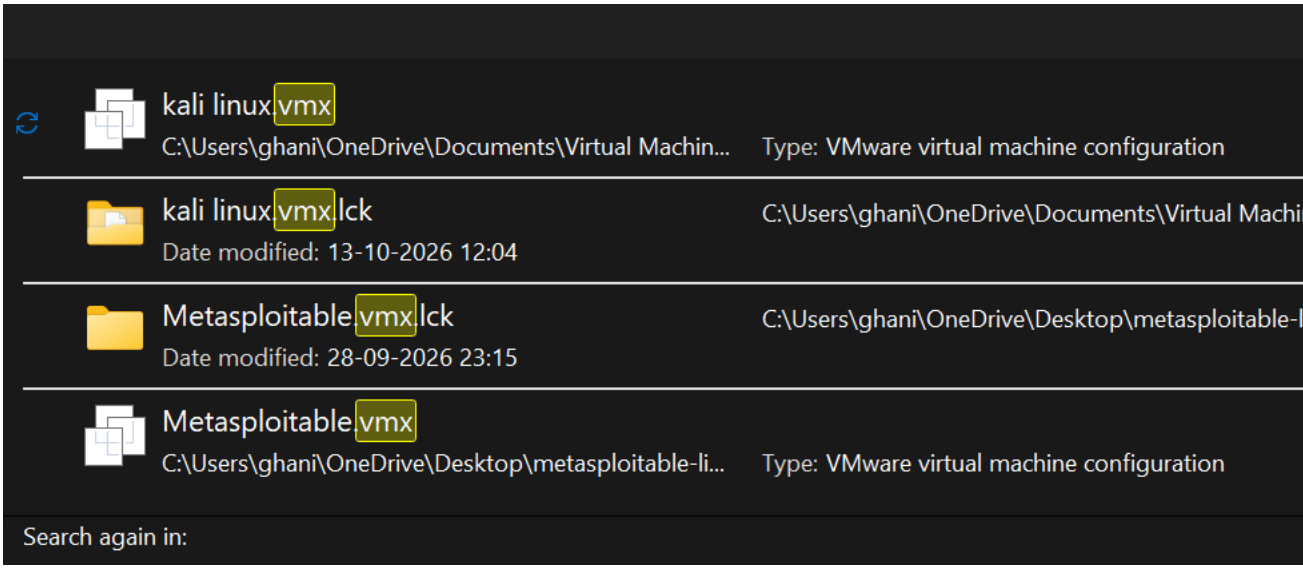
3.1 Lab Setup

VMware Workstation was used to manage the virtual machine environment. The available systems included Kali Linux and Metasploitable 2.

Kali Linux was used for Linux command-line practice and network configuration checks. Metasploitable 2 was identified as the intentionally vulnerable target system for controlled security practice.

The lab work also involved understanding NAT, bridged, and host-only networking modes.

Safety consideration: Metasploitable 2 should be kept isolated from public and untrusted networks. Security testing should be performed only in an authorized environment.



```
┌──(ghanisth㉿ghanisth)-[~]
└─$ whoami
ghanisth

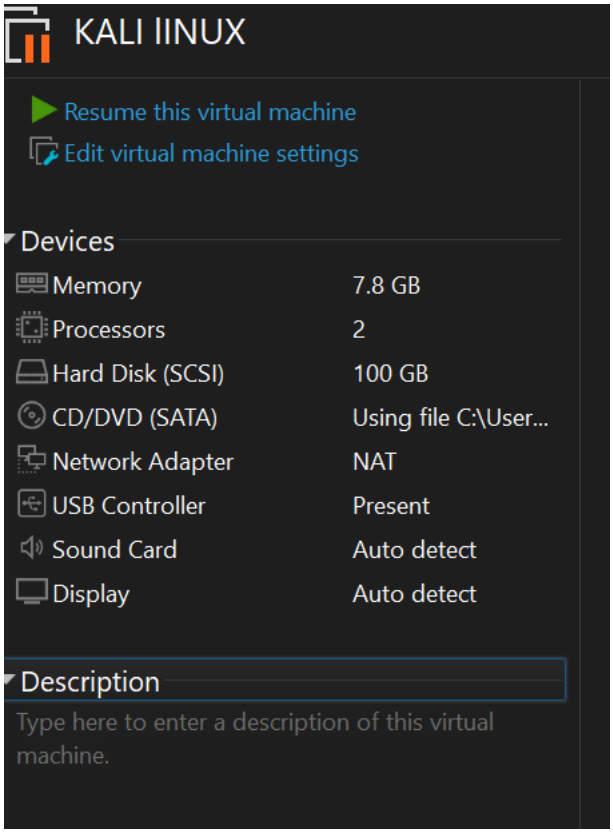
┌──(ghanisth㉿ghanisth)-[~]
└─$ hostname
ghanisth

┌──(ghanisth㉿ghanisth)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:59:bb:67 brd ff:ff:ff:ff:ff:ff
    inet 192.168.144.128/24 brd 192.168.144.255 scope global dynamic noprefixroute eth0
       valid_lft 1766sec preferred_lft 1766sec
    inet6 fe80::20c:29ff:fe59:bb67/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 5a:1a:85:3a:c7:46 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
    inet6 fe80::581a:85ff:fe3a:c746/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
14: veth68a26e7@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether 86:1e:d6:1f:5d:3e brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::841e:d6ff:fe1f:5d3e/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
17: br-14e8bf5c2d44: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether ba:2e:bc:d1:d1:3e brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-14e8bf5c2d44
       valid_lft forever preferred_lft forever
    inet6 fe80::b82e:bcff:fed1:d13e/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
18: vethd9bba50@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-14e8bf5c2d44 state UP group default
    link/ether fa:28:74:b6:af:0b brd ff:ff:ff:ff:ff:ff link-netnsid 1
    inet6 fe80::f828:74ff:feb6:af0b/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
19: vethc686316@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-14e8bf5c2d44 state UP group default
    link/ether 3a:b7:32:45:94:f4 brd ff:ff:ff:ff:ff:ff link-netnsid 2
    inet6 fe80::38b7:32ff:fe45:94f4/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever

┌──(ghanisth㉿ghanisth)-[~]
└─$ ip route
default via 192.168.144.2 dev eth0 proto dhcp src 192.168.144.128 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1
172.18.0.0/16 dev br-14e8bf5c2d44 proto kernel scope link src 172.18.0.1
192.168.144.0/24 dev eth0 proto kernel scope link src 192.168.144.128 metric 100
```

```
msfadmin@metasploitable:~$
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:79:a1:18
          inet addr:192.168.144.129  Bcast:192.168.144.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe79:a118/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:77264 errors:0 dropped:0 overruns:0 frame:0
          TX packets:61493 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:12652249 (12.0 MB)  TX bytes:23773010 (22.6 MB)
          Interrupt:17 Base address:0x2000

msfadmin@metasploitable:~$
```



Use your actual screenshots from `Findings/Lab Setup/`. If a screenshot is unavailable, omit it rather than claiming that it was captured.

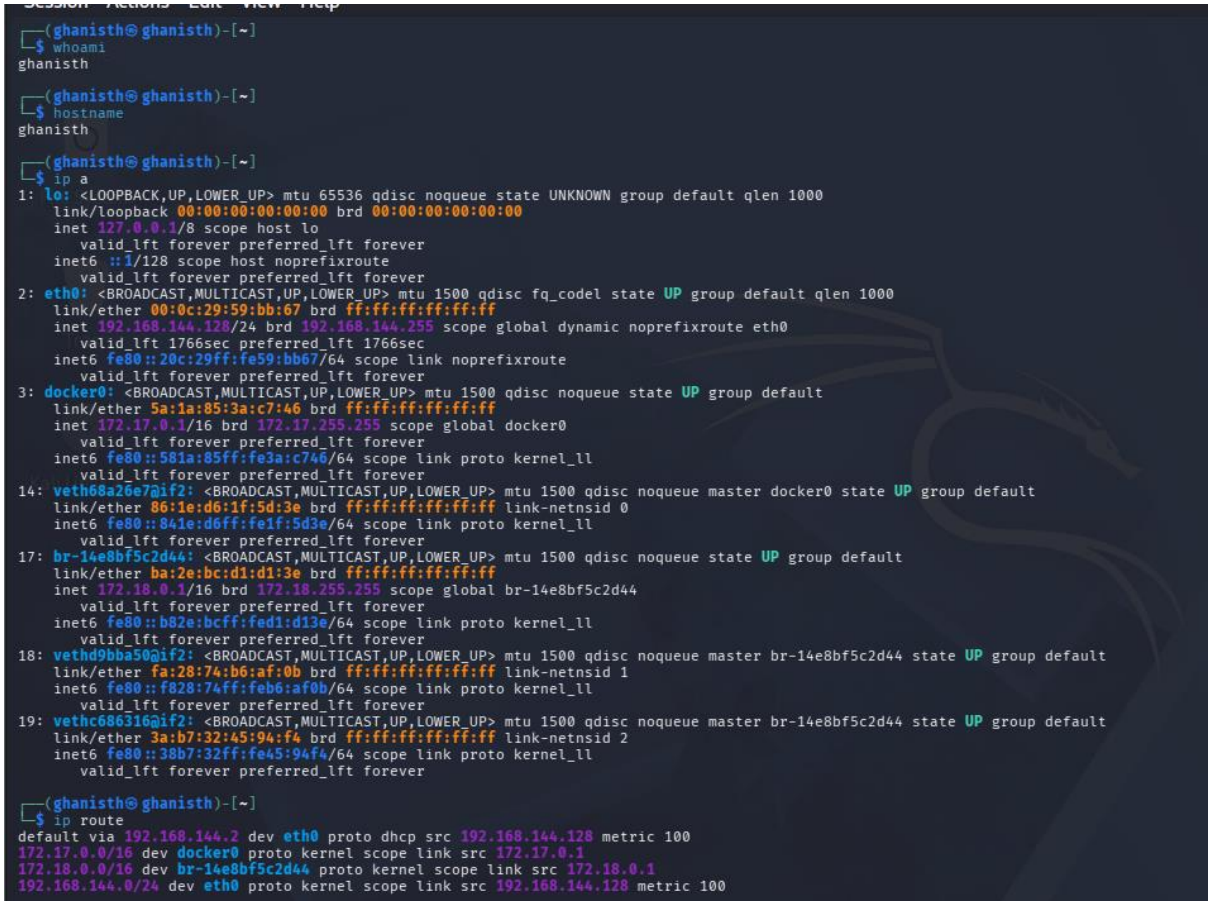
3.2 Network Configuration

The network configuration of Kali Linux was inspected using Linux networking commands. These commands helped identify the available network interfaces, IP addresses, default gateway, and DNS configuration.

The following commands were used:

```
ip a
ip route
cat /etc/resolv.conf
```

These commands are useful for troubleshooting connectivity and understanding how a system communicates with other networks.



4. LINUX COMMAND-LINE PRACTICE

4.1 Basic Linux Commands

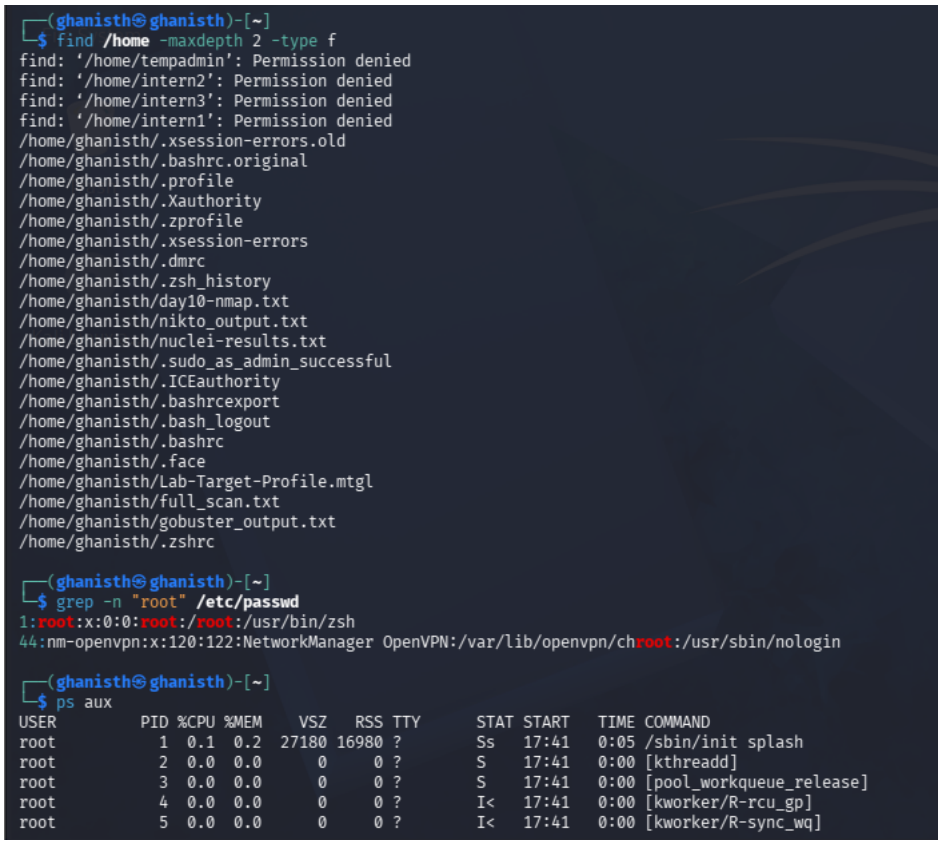
Kali Linux was used to practice commands for system information, directory navigation, file searching, network inspection, and process management.

The commands practiced included:

Command	Purpose
`whoami`	Displays the current username
`hostname`	Displays the system hostname
`pwd`	Shows the current directory
`ls -la`	Lists files, including hidden files
`cat /etc/os-release`	Displays operating system information
`find`	Searches for files and directories
`grep`	Searches for matching text
`ps aux`	Displays running processes
`ip a`	Displays network interfaces and addresses
`ip route`	Displays routing information

`sudo ss -tulpn`	Inspects listening sockets and associated processes where permitted
`chmod`	Changes file permissions
`crontab -e`	Edits the current user's scheduled tasks

These commands provide a foundation for Linux administration and cybersecurity investigations.



4.2 Users, Groups, and File Permissions

An important practical exercise involved creating users and groups and configuring shared directory access.

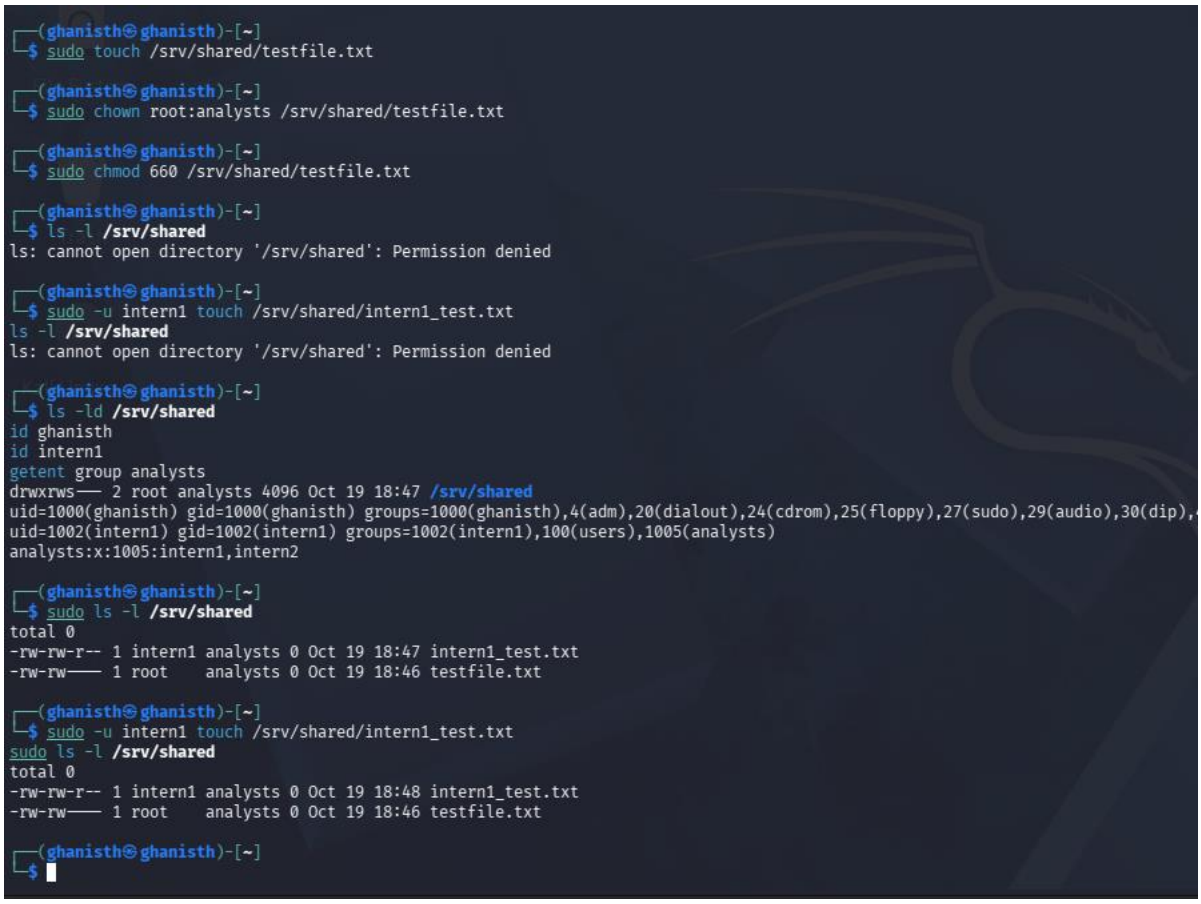
The users `intern1`, `intern2`, and `intern3` were used during the exercise. The groups `analysts` and `testers` were created to demonstrate group-based access control.

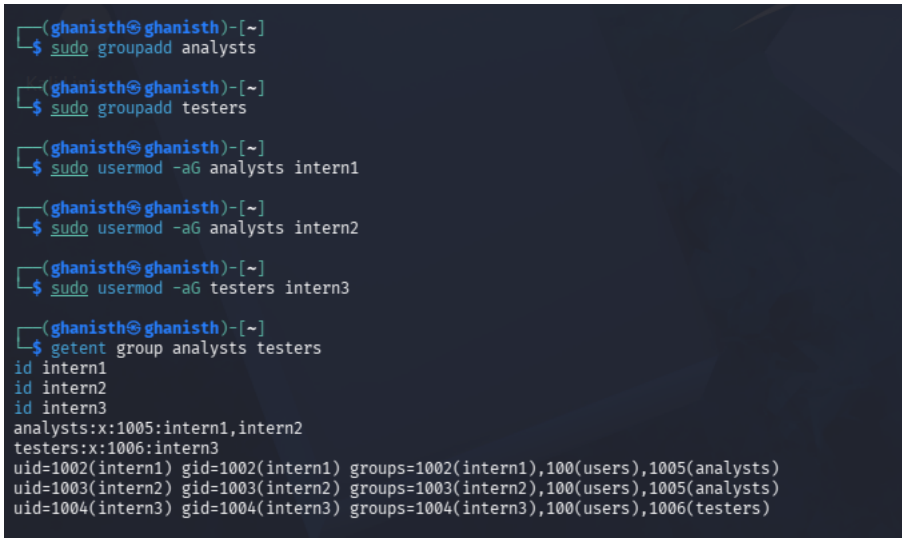
The shared directory `/srv/shared` was configured with group ownership and restricted permissions. The setgid permission was used so that files created within the directory could inherit its group ownership.

The permission mode `2770` allows the directory owner and group members to access the directory according to the assigned permissions while denying access to other users.

A permission-denied result for a user outside the authorized group demonstrated that access restrictions were working as expected.

This exercise showed why access should be granted according to user roles instead of giving every user unrestricted permissions.





4.3 Cron Job Configuration

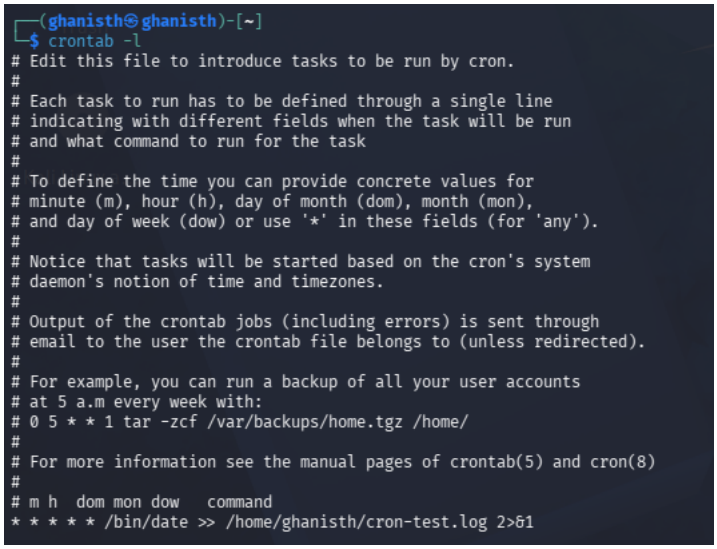
A cron job was configured to run a command periodically and append the current date and time to a log file.

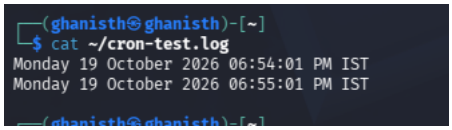
Example command:

```
* * * * * /bin/date >> /home/ghanisth/cron-test.log 2>&1
```

The expression schedules the command to run once every minute. The output is appended to the specified log file.

This exercise introduced task scheduling and basic log verification in Linux.





5. NETWORKING AND CONNECTIVITY TESTING

5.1 Network Commands

Networking exercises were performed to understand IP configuration, DNS resolution, connectivity, and the route taken by network traffic.

The following commands were used:

```
ip a
```

```
ip route
cat /etc/resolv.conf
ping -c 4 8.8.8.8
ping -c 4 google.com
nslookup google.com
traceroute google.com
```

The commands served the following purposes:

- `ip a`: Inspected network interfaces and assigned IP addresses.
- `ip route`: Identified routing information and the default gateway.
- `cat /etc/resolv.conf`: Displayed configured DNS resolver information.
- `ping`: Tested connectivity to an IP address or hostname.
- `nslookup`: Checked DNS name resolution.
- `traceroute`: Examined the network path toward a destination.

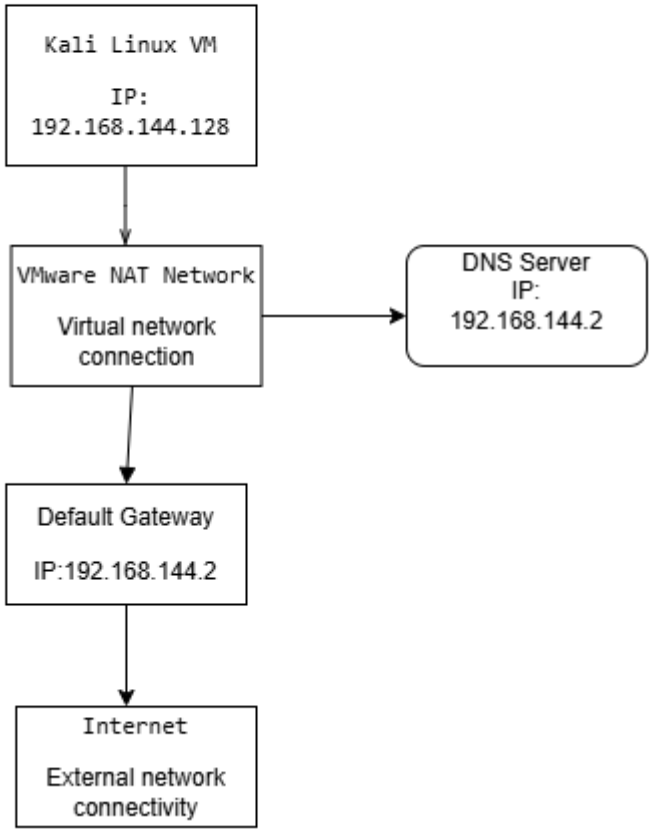
The results helped distinguish IP connectivity from DNS resolution and provided a basic understanding of network troubleshooting.

5.2 Network Map

A network map was prepared to represent the Kali Linux virtual machine, its virtual network connection, the gateway, DNS resolution, and the internet connectivity path.

The diagram provides a visual summary of the observed lab network.

Kali Linux Network Map



```
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:59:bb:67 brd ff:ff:ff:ff:ff:ff
    inet 192.168.144.128/24 brd 192.168.144.255 scope global dynamic noprefixroute eth0
       valid_lft 1494sec preferred_lft 1494sec
    inet6 fe80::20c:29ff:fe59:bb67/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: br-14e8bf5c2d44: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether d2:68:5a:18:2d:7a brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-14e8bf5c2d44
       valid_lft forever preferred_lft forever
    inet6 fe80::d068:5aff:fe18:2d7a/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 96:35:8d:5c:45:41 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
5: veth98c349a@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-14e8bf5c2d44 state UP group default
    link/ether 9a:da:e8:35:6e:fa brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::98da:e8ff:fe35:6efa/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
6: vethf3391b5@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-14e8bf5c2d44 state UP group default
    link/ether 1e:39:e0:98:2f:da brd ff:ff:ff:ff:ff:ff link-netnsid 1
    inet6 fe80::1c39:e0ff:fe98:2fda/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever

┌──(ghanisth㉿ghanisth)-[~]
└─$ ip route
default via 192.168.144.2 dev eth0 proto dhcp src 192.168.144.128 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
172.18.0.0/16 dev br-14e8bf5c2d44 proto kernel scope link src 172.18.0.1
192.168.144.0/24 dev eth0 proto kernel scope link src 192.168.144.128 metric 100

┌──(ghanisth㉿ghanisth)-[~]
└─$ cat /etc/resolv.conf
# Generated by NetworkManager
search localdomain
nameserver 192.168.144.2
```

```
┌──(ghanisth㉿ghanisth)-[~]
└─$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=14.7 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=11.2 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=13.2 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=13.8 ms

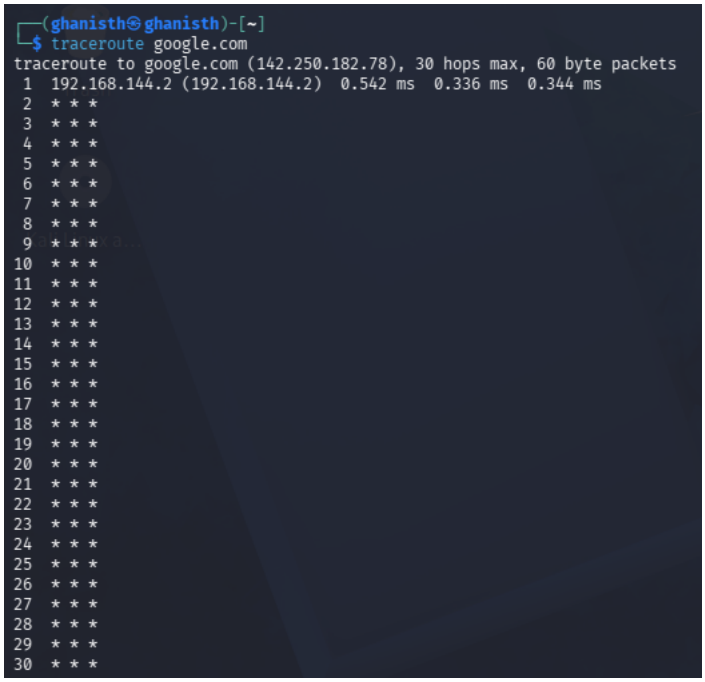
── 8.8.8.8 ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 11.159/13.210/14.739/1.309 ms

┌──(ghanisth㉿ghanisth)-[~]
└─$ ping -c 4 google.com
PING google.com (142.250.182.78) 56(84) bytes of data.
64 bytes from maa05s20-in-f14.1e100.net (142.250.182.78): icmp_seq=1 ttl=128 time=52.5 ms
64 bytes from maa05s20-in-f14.1e100.net (142.250.182.78): icmp_seq=2 ttl=128 time=56.0 ms
64 bytes from maa05s20-in-f14.1e100.net (142.250.182.78): icmp_seq=3 ttl=128 time=59.2 ms
64 bytes from maa05s20-in-f14.1e100.net (142.250.182.78): icmp_seq=4 ttl=128 time=53.7 ms

── google.com ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 52.468/55.330/59.163/2.558 ms

┌──(ghanisth㉿ghanisth)-[~]
└─$ nslookup google.com
Server:         192.168.144.2
Address:        192.168.144.2#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.182.78
Name:   google.com
Address: 2404:6800:4013:807::71
Name:   google.com
Address: 2404:6800:4013:807::8b
Name:   google.com
Address: 2404:6800:4013:807::65
Name:   google.com
Address: 2404:6800:4013:807::8a
```



Evidence folder: `Findings/Network Map - Diagram/`

The IP address, gateway, and DNS server recorded in the report should match the actual command outputs and network diagram.

6. OVER THE WIRE BANDIT

OverTheWire Bandit was used to practice Linux command-line skills through progressively challenging exercises. Levels 0–10 introduced file handling, hidden files, searching, text processing, and encoding.

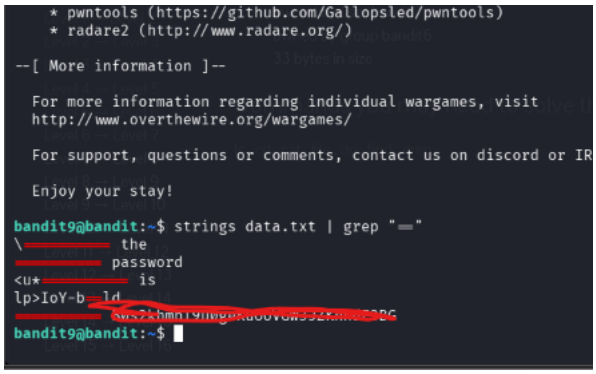
Level	Concept practiced
0	Connecting through SSH and reading a file
1	Reading a file with a special-character filename
2	Handling filenames containing spaces
3	Locating hidden files
4	Identifying file types
5	Searching files by properties
6	Searching by owner, group, and file size
7	Searching file contents using `grep`
8	Identifying a unique line using `sort` and `uniq`
9	Extracting readable strings
10	Decoding Base64 data

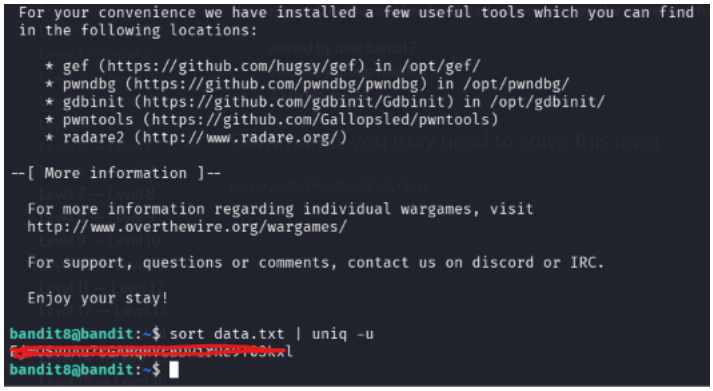
The exercises improved familiarity with terminal-based problem solving and common Linux utilities.

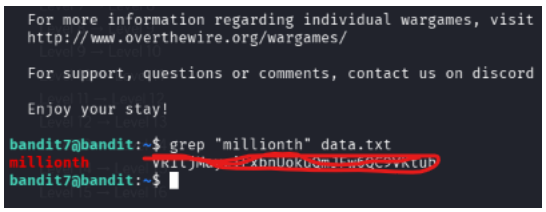
A separate write-up was prepared to document the concepts and commands used during the exercises.

If the report becomes too long, include a few representative screenshots in this section and keep the remaining screenshots in the evidence folder.

Evidence folder: `Findings/Over the Wire Bandit/`









```
bandit5@bandit:~$ cd inhere
bandit5@bandit:~/inhere$ find . -type f -size 1033c
./maybehere07/.file2
bandit5@bandit:~/inhere$ cat ./maybehere07/.file2
[redacted]
                                                          bandit5@bandit:~/inhere$ exit
logout
Connection to bandit.labs.overthewire.org closed.
```

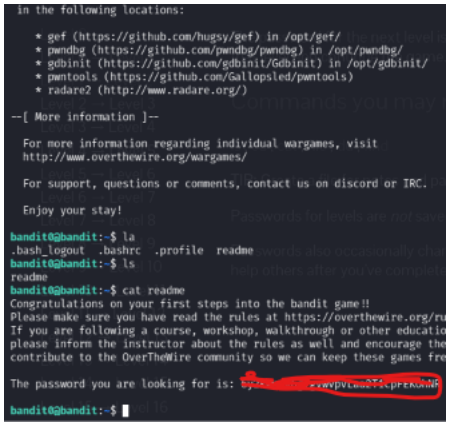
```
bandit4@bandit:~$
bandit4@bandit:~$ cd inhere
bandit4@bandit:~/inhere$ ls -la
total 48
-rw-r----- 1 bandit5 bandit4   33 Sep 26 21:54 -file00
-rw-r----- 1 bandit5 bandit4   33 Sep 26 21:54 -file01
-rw-r----- 1 bandit5 bandit4   33 Sep 26 21:54 -file02
-rw-r----- 1 bandit5 bandit4   33 Sep 26 21:54 -file03
-rw-r----- 1 bandit5 bandit4   33 Sep 26 21:54 -file04
-rw-r----- 1 bandit5 bandit4   33 Sep 26 21:54 -file05
-rw-r----- 1 bandit5 bandit4   33 Sep 26 21:54 -file06
-rw-r----- 1 bandit5 bandit4   33 Sep 26 21:54 -file07
-rw-r----- 1 bandit5 bandit4   33 Sep 26 21:54 -file08
-rw-r----- 1 bandit5 bandit4   33 Sep 26 21:54 -file09
drwxr-xr-x 2 root    root    4096 Sep 26 21:54 .
drwxr-xr-x 3 root    root    4096 Sep 26 21:54 ..
bandit4@bandit:~/inhere$ file ./*
./-file00: data
./-file01: data
./-file02: data
./-file03: data
./-file04: data
./-file05: data
./-file06: data
./-file07: ASCII text
./-file08: data
./-file09: data
bandit4@bandit:~/inhere$ cat ./-file07
[redacted]
bandit4@bandit:~/inhere$
```

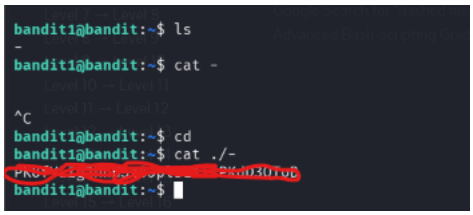
```
bandit3@bandit:~$ ls
inhere
bandit3@bandit:~$ cd inhere
bandit3@bandit:~/inhere$ ls
bandit3@bandit:~/inhere$ ls -la
total 12
drwxr-xr-x 2 root    root    4096 Sep 26 21:54 .
drwxr-xr-x 3 root    root    4096 Sep 26 21:54 ..
-rw-r----- 1 bandit4 bandit3   33 Sep 26 21:54 ...Hiding-From-You
bandit3@bandit:~/inhere$ cat .hidden
cat: .hidden: No such file or directory
bandit3@bandit:~/inhere$ cat ...Hiding-From-You
[redacted]4lfQytTWufAMq
bandit3@bandit:~/inhere$
```

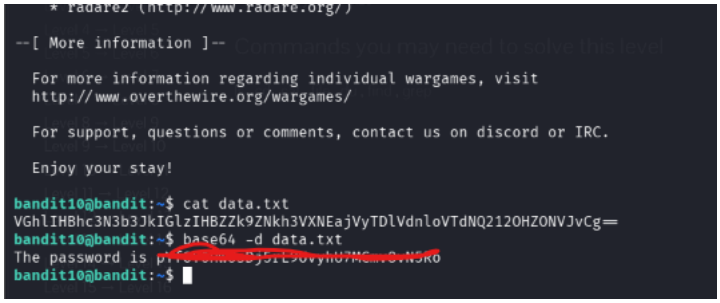
```
  For support, questions or comments, contact us on discord or

  Enjoy your stay!

bandit2@bandit:~$ ls
--spaces in this filename--
bandit2@bandit:~$ cat ./"--spaces in this filename--"
[redacted]
bandit2@bandit:~$
```







No passwords or challenge credentials should be displayed in the report.

7. TRYHACKME LEARNING ACTIVITIES

Eight beginner-level TryHackMe rooms were included in the Week 1 learning plan.

No.	Room	Learning focus
1	Search Skills	Searching for information and useful resources
2	Learning Cyber Security	Cybersecurity roles and fundamental concepts
3	Linux Fundamentals Part 1	Basic Linux commands and file navigation
4	What Is Networking?	Networking fundamentals
5	Introductory Networking	Basic networking concepts and protocols
6	Networking Concepts	Network communication and related concepts
7	Pyramid of Pain	Indicators of compromise and detection difficulty
8	Intro to LAN	Local area networks and network devices

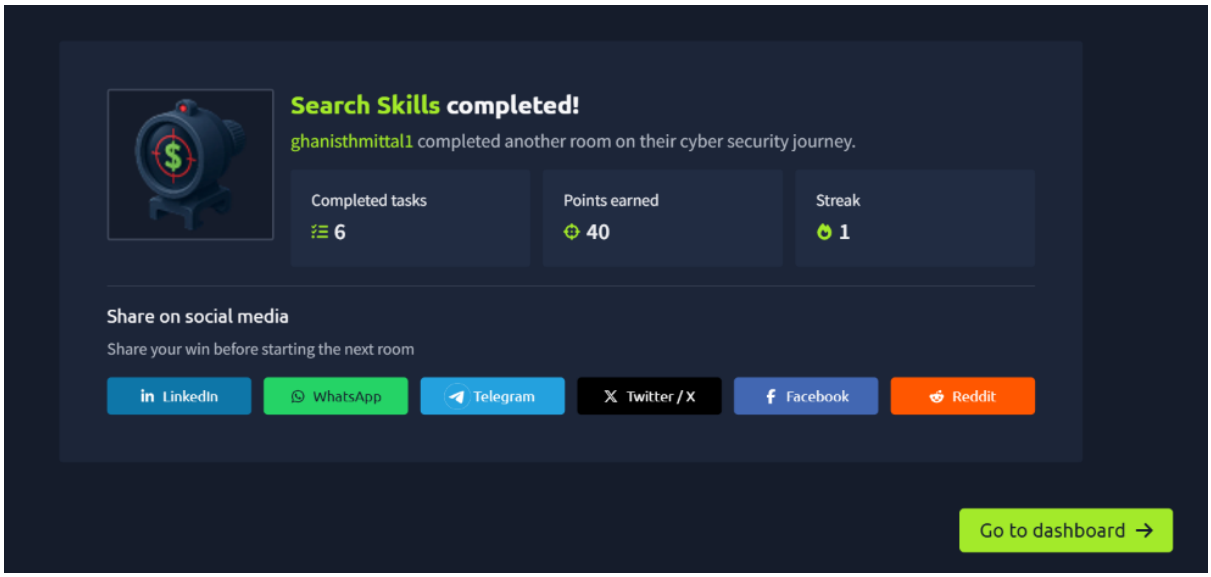
These rooms provided guided exercises to reinforce the theoretical concepts studied during the week.

Insert screenshots here

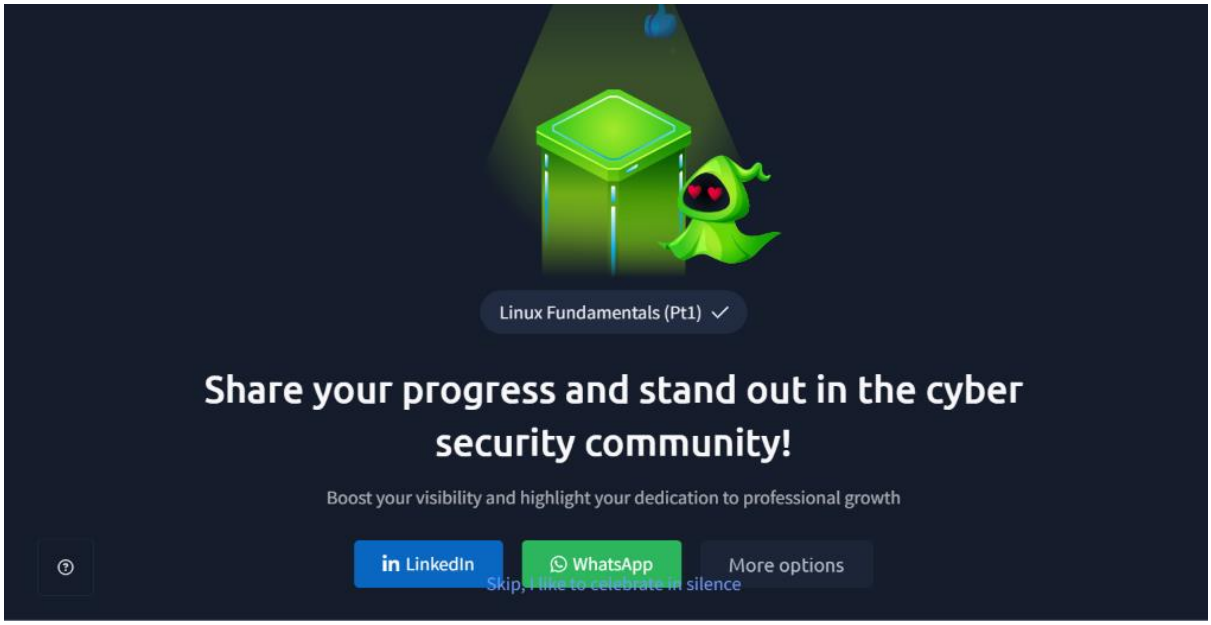
Insert the completion screenshot for each room after its corresponding room description, or place all eight screenshots together in an appendix.

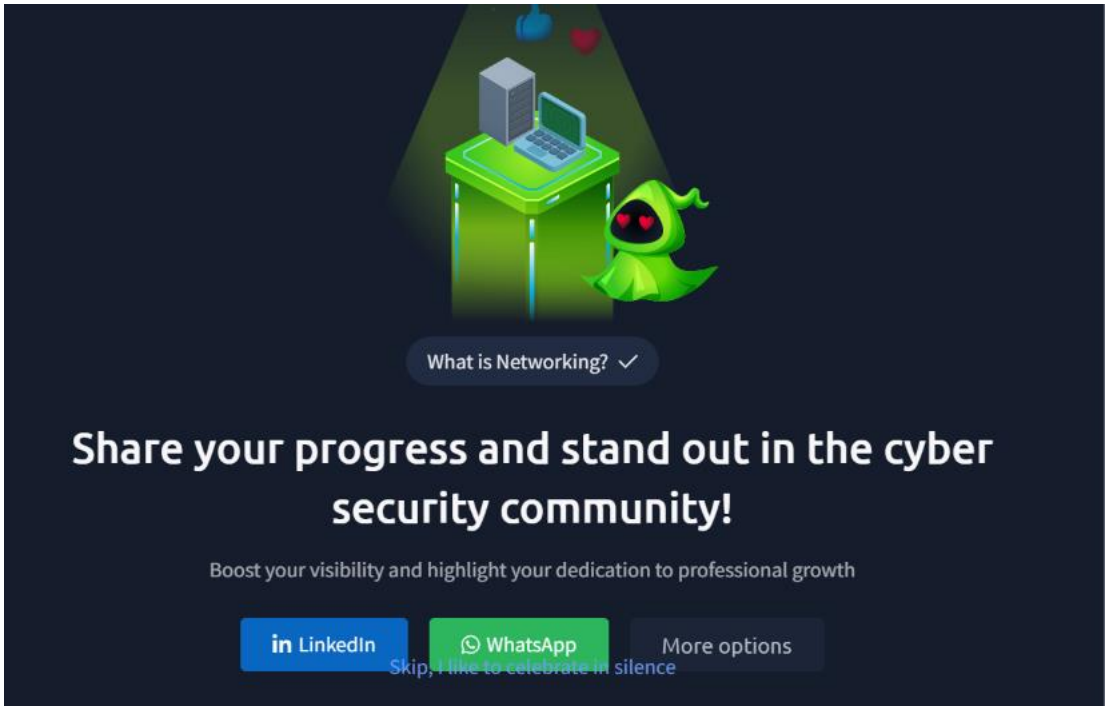
Use these filenames:

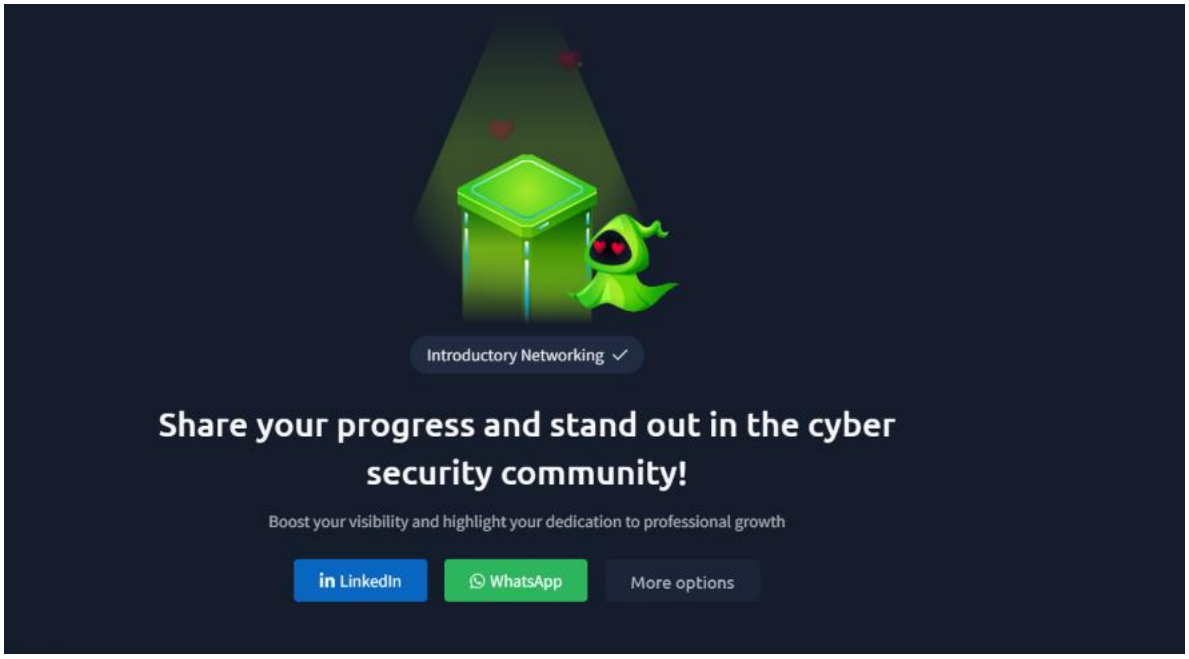
Evidence folder: `Findings/TryHackMe/`

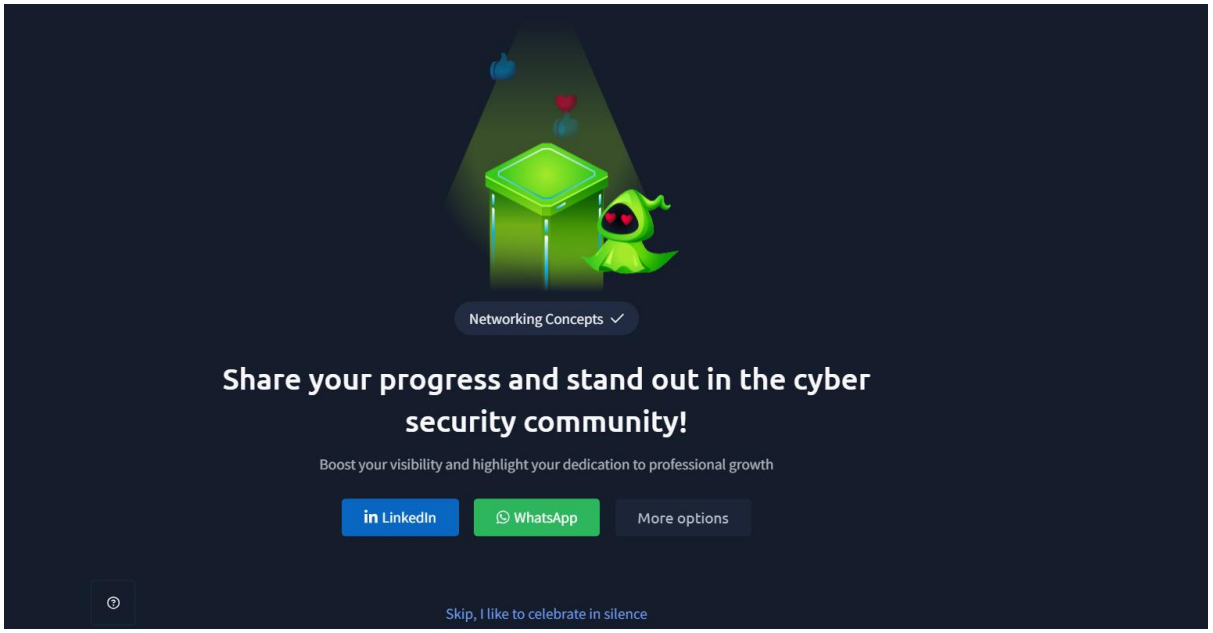


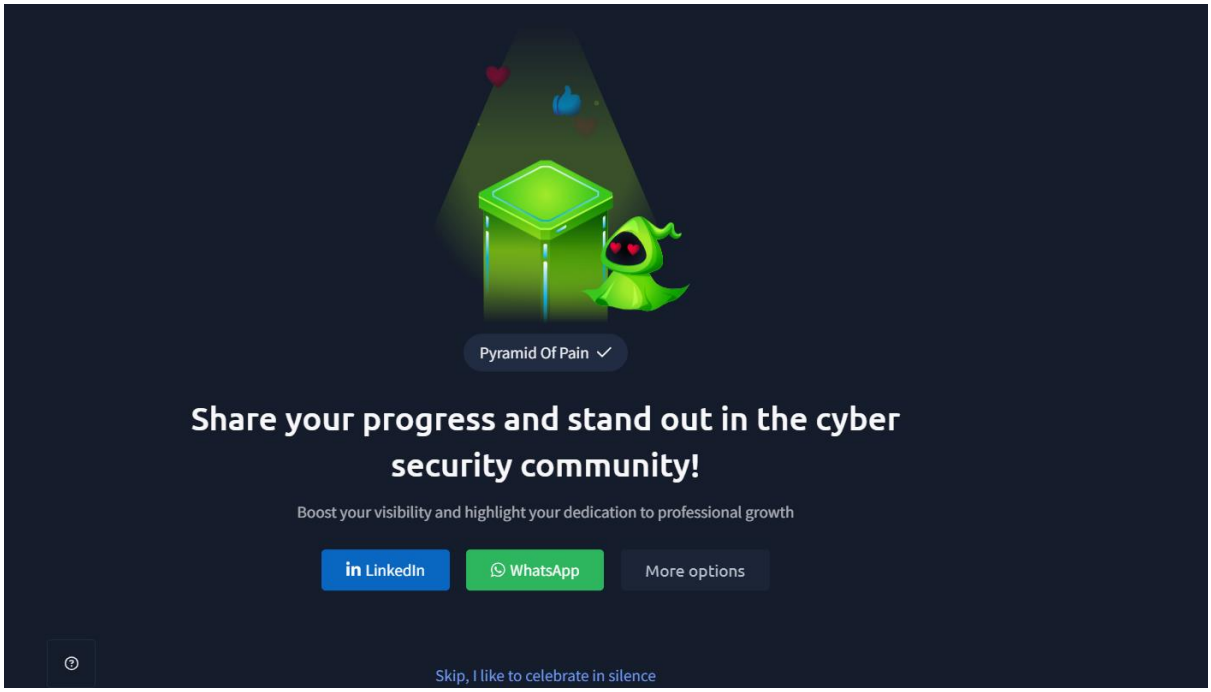


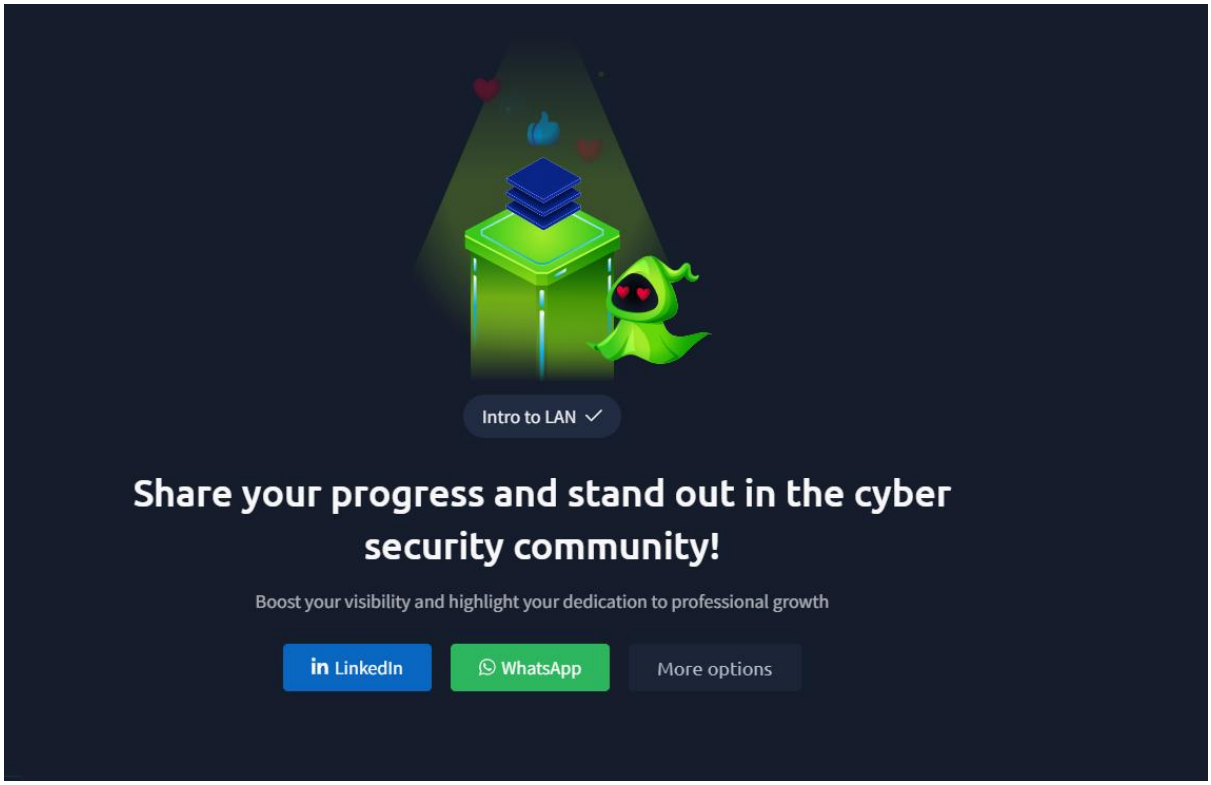

Linux Fundamentals (Pt1)
Share your progress and stand out in the cyber security community!
Boost your visibility and highlight your dedication to professional growth
LinkedIn
WhatsApp
More options
Skip, I like to celebrate in silence


What is Networking?
Share your progress and stand out in the cyber security community!
Boost your visibility and highlight your dedication to professional growth
LinkedIn
WhatsApp
More options
Skip, I like to celebrate in silence


Introductory Networking
Share your progress and stand out in the cyber security community!
Boost your visibility and highlight your dedication to professional growth
LinkedIn
WhatsApp
More options


Networking Concepts
Share your progress and stand out in the cyber security community!
Boost your visibility and highlight your dedication to professional growth
LinkedIn
WhatsApp
More options
Skip, I like to celebrate in silence





Each screenshot should clearly show the corresponding room and its completion status. The folder name identifies which room the screenshot belongs to.

8. LINUX COMMAND CHEAT SHEET

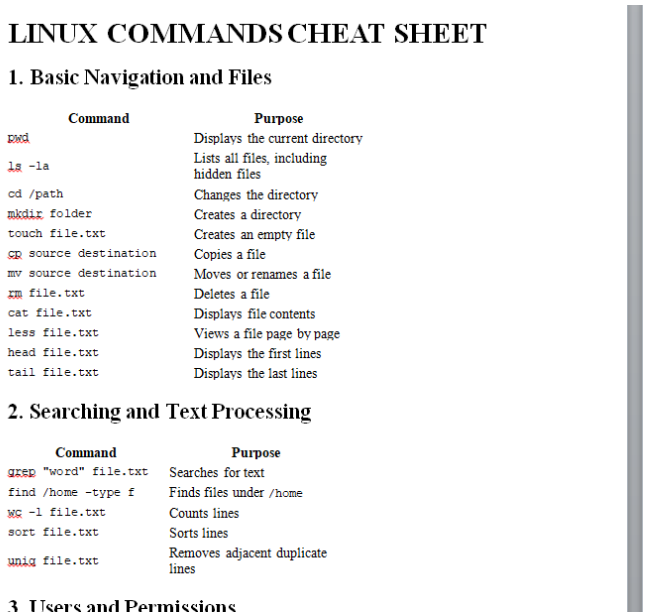
A separate Linux command cheat sheet was prepared for quick reference during practical work.

It covers common commands for:

- File and directory management.
- System information.
- Searching and filtering text.
- User and group administration.
- File permissions.
- Process inspection.
- Network configuration and troubleshooting.
- Scheduled tasks.

The cheat sheet is intended to support future Linux and cybersecurity practical exercises.

Document: `Cheat Sheet/Linux_Commands_Cheat_Sheet.docx`



LINUX COMMANDS CHEAT SHEET

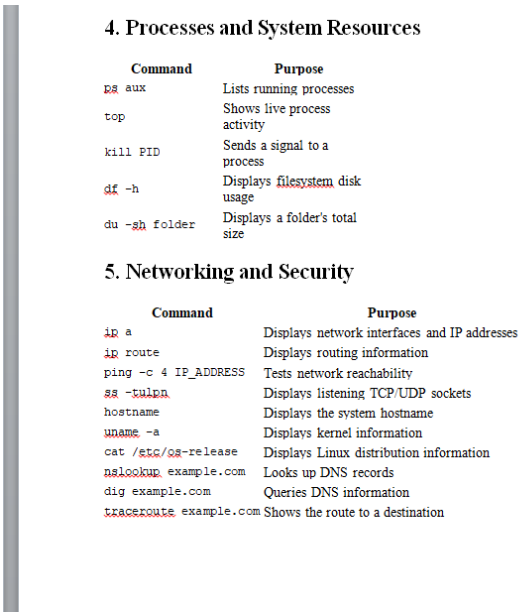
1. Basic Navigation and Files

Command	Purpose
pwd	Displays the current directory
ls -la	Lists all files, including hidden files
cd /path	Changes the directory
mkdir folder	Creates a directory
touch file.txt	Creates an empty file
cp source destination	Copies a file
mv source destination	Moves or renames a file
rm file.txt	Deletes a file
cat file.txt	Displays file contents
less file.txt	Views a file page by page
head file.txt	Displays the first lines
tail file.txt	Displays the last lines

2. Searching and Text Processing

Command	Purpose
grep "word" file.txt	Searches for text
find /home -type f	Finds files under /home
wc -l file.txt	Counts lines
sort file.txt	Sorts lines
uniq file.txt	Removes adjacent duplicate lines

3. Users and Permissions



4. Processes and System Resources

Command	Purpose
ps aux	Lists running processes
top	Shows live process activity
kill PID	Sends a signal to a process
df -h	Displays filesystem disk usage
du -sh folder	Displays a folder's total size

5. Networking and Security

Command	Purpose
ip a	Displays network interfaces and IP addresses
ip route	Displays routing information
ping -c 4 IP_ADDRESS	Tests network reachability
ss -tulpn	Displays listening TCP/UDP sockets
hostname	Displays the system hostname
uname -a	Displays kernel information
cat /etc/os-release	Displays Linux distribution information
nslookup example.com	Looks up DNS records
dig example.com	Queries DNS information
traceroute example.com	Shows the route to a destination

9. KEY LEARNING OUTCOMES

The activities undertaken during Week 1 helped develop the following foundational skills:

1. **Cybersecurity fundamentals:** Improved understanding of confidentiality, integrity, availability, threats, vulnerabilities, and risk.
2. **Linux administration:** Practiced basic commands, file searches, permissions, user management, and group management.
3. **Access control:** Learned how group membership and directory permissions control access to shared resources.
4. **Networking:** Practiced inspecting IP addresses, routing, DNS configuration, and connectivity.

5. **Virtual labs:** Learned about virtual machine networking modes and the importance of isolating vulnerable targets.
6. **Problem-solving:** Used command-line tools to investigate files and solve beginner-level challenges.
7. **Technical documentation:** Organized notes, diagrams, screenshots, and write-ups into structured folders.
8. **Ethical practice:** Reinforced the importance of authorization and safe testing environments.

10. CHALLENGES AND SAFETY CONSIDERATIONS

Practical work required attention to file permissions, user access, network configuration, and correct interpretation of command output.

Permission-denied messages helped demonstrate the importance of access control. Network testing also highlighted the need to check IP connectivity and DNS resolution separately.

All practical testing should remain within authorized environments. Vulnerable systems should be isolated, credentials should be kept private, and screenshots should be checked for sensitive information before submission.

Any task that was not completed or could not be verified should be identified honestly in the final submission.

Report prepared for: DG Interns Hub — Week 1 Cybersecurity Internship

Submitted by: Ghanisth Mittal